



Wear Cap Installation Tool for ERS 7495

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

In the routine activity of removing and installing a Wear Cap - WC 62 (PN398-3042) for the ERS 7495 Dipper lip adapter, an accident occurred involving an employee's hand and fingers. It was observed that the activity of replacing the WC to protect the penetration adapters presents ergonomic risks related to hand and finger crushing. The activity was performed manually by only one employee. This incident was widely publicized among the leaders of the client Vale S11D and Sotreq, as shown in Figure 1 below.

GERÊNCIA: GER MANUTENÇÃO DE EQUIPAMENTO MÓVEIS
 SUPERVISÃO: SUP PREVENTIVA ESCAVAÇÃO – HELDER RODRIGUES
 TÍTULO DO KAIZEN: Suporte para instalação de (FPS) – EC 54

PROBLEMA: ☐ Transporte ☐ Espera ☐ Estoque ☐ Superprodução ☐ Processamento Excessivo ☐ Defeito ☒ Movimento ☐ MURI ☐ MURA ☐ 5S

Estado anterior

Após um evento N4 envolvendo mãos e dedos foi observado que a atividade de substituição dos capacetes de proteção da caçamba (FPS), apresentavam fatores ergonômicos, tais como: manuseio ou movimentação de cargas e volumes sem prego ou com "pega pobre"; trabalho em posturas incômodas ou pouco confortáveis; exigência de flexões de coluna vertebral frequentes e riscos de esmagamentos de mãos e dedos. A atividade é realizada manualmente apenas por um colaborador.

Estado atual

Elaborado uma forma mais Segura para a execução da atividade, com essa melhoria pode ser evitado a exposição direta de um colaborador evitando que o mesmo realize a atividade com a capacidade de carga individual acima do permitido que são 20kg por empregado, e a capacidade do capacete é de 27kg.

Benefícios do processo	ID da melhoria	ID GMUD
SSMAC & RISCOS Reduções dos fatores ergonômicos na realização da atividade, eliminando riscos de prensamento de mãos e dedos devido o colaborador não ter contato direto com o capacete.	dbc7aea7-60d0-4f97-bc51-9c4407178e87	N/A
PRODUTIVIDADE N/A	CUSTOS N/A	QUALIDADE/SS Atividade realizada com menor exposição ao risco de incidentes e com reduções de fatores ergonômicos ao realizar a mesma.

PADRONIZAÇÃO: Nº do padrão: N/A

Figure 1: Installation Procedure Comparison

The Wear Cap used at the time is the WC-62 with 28kg. Dimensions are described in Figure 3 below.

Adapters											Notes
Type	Description	Part #	Wear Cap	A (mm)	B (mm)	C (mm)	D (mm)	E (mm)	F °	Wt (kg)	
 WC-62 WC-62LWP	WC-45	398-3040	A	301	210	–	45	–		20	 A B D
	WC-62	398-3042	A	301	210	–	62	–		28	
	WC-62LWP	503-1306	A	301	210	–	77	–		36	
	WC-53	398-3041	B	276	192	–	53	–		27	
	WC-53LWP	598-0634	B	276	192	–	79	–		34	

Figure 2: Wear Cap Information

A method was developed to mitigate the possibility of future accidents. The solution is a magnetic base and support with a lifting capacity of 400kg that could be used by two people more safely.

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2.0 Best Practice Description

The idea of this best practice was to provide a safe way of removing and installing Wear Caps.

A method and structure was developed that allows workers to avoid moving the Wear Cap directly with his hands and at the same time mitigate ergonomic damage by dividing the hoisted load between two people.

3.0 Implementation Steps

The implementation of this best practice can be divided into 3 steps, outlined below.

STEP 1: Determine Magnet Based on Lifting Requirements

The magnetic base has:

- 1) The capacity for lifting plates weighing up to 400kg
- 2) A safety factor of 3.5
- 3) An operating temperature of up to 80°C.



Figure 3: Magnet

The dimension of the chosen magnet are 1) a height of 96mm, 2) a width of 94mm and 3) a length of 168mm. It does not need electrical power as it is built with high energy permanent magnets.

STEP 2: Build Lifting Structure

The support is made of 1/2" ABNT 1020 steel tubes for greater strength. It has two 500mm lifting rods and a 314.9 mm "V". Using two technicians, the support is coupled to the magnetic base and the lifting is carried out by them.

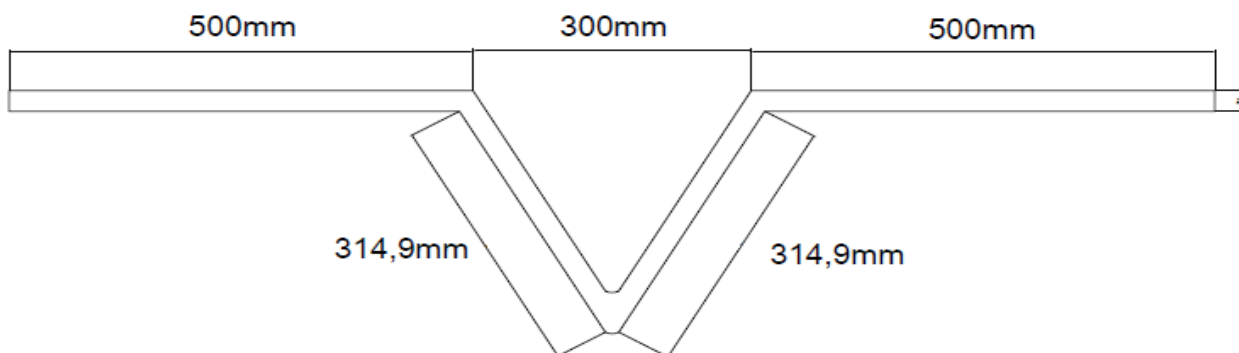


Figure 4: Support Description

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STEP 3: How to Use

The lifting tool developed specifically for this best practice is inserted into the upper eye of the magnetic lifter. A technician performs the fitting using a rod to pull the wear cap as shown in Figure 6 below. Using two technicians, the dipper adapter wear cap can be lifted.



Figure 5: Lift Execution

4.0 Benefits

For the Wear Cap R&I:

- Greater security in carrying out the R&I activities
- Decreased risk of crushing and pinching fingers

5.0 Resources Required

- 1 magnetic base with a capacity of 400kg with eyelet
- 1 lifting rod with two handles
- 1 rod to pull the helmet already attached to the magnetic lifter

6.0 Supporting Attachments / References

Cat Ground Engaging Tool – Parts Catalog: HMS, ERS, DGL & LHD

7.0 Related Best Practices

N/A

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8.0 Acknowledgements

We thank the entire engineering team that was involved in the development of this best practice:

Thiago Dillion

Expanded Mining Products

thiago.dillon@sotreq.com.br

[+55 94 9](tel:+55949)

Anderson Lima

Expanded Mining Products

anderson.lima@sotreq.com.br

Webert Lopes

Expanded Mining Products

webert.lopes@sotreq.com.br

Thiago Rezende

Expanded Mining Products

thiago.rezende@sotreq.com.br

Rudson Maciel

Expanded Mining Products

rudson.maciel@sotreq.com.br

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